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Wildfire risk modeling

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Highlights

- Risk assessment and modeling are crucial to understand wildfire drivers and impacts.
- Methods and tools used to model key components of wildfire risk are described.
- Recent approaches and challenges in wildfire modeling are indicated.

Abstract

Wildfires occur worldwide and affect different territories in diverse ways. New challenges in wildfire management are emerging, fuelled by climatic, land cover, and sociodemographic changes. Fostering wildfire science to respond with advanced and inclusive approaches. Supported by technological improvements, wildfire risk modeling is earning relevance in several wildfire phases, covering prevention, suppression, and recovery, and encouraging the development of different products and tools. Most wildfire risk approaches reflect a broad conceptual framework that includes different but interconnected components, which can also be analyzed separately, such as likelihood, hazard, exposure, vulnerability, and coping capacity. This article presents a concise and nonexhaustive review of the methods and approaches applied to model wildfire risk and its key components, in particular since 2018, and how they can contribute to wildfire management.

Introduction

Wildfires occur globally and affect different territories in distinct ways. In the last decades, the incidence of large and uncontrolled fires has risen, and extreme events are expected to become more frequent and destructive [1]. Exceptionally large and damaging fires have recently occurred, for example, in California in 2017 in the western USA in 2018, and in Brazil and eastern Australia in 2019 [2, 3, 4, 5]. Changes in land cover, climate, and sociodemographic conditions are pointed out as the main reasons behind these unprecedented events [6], but further understanding is needed on how these changes drive wildfire risk in different spatial contexts and how this knowledge can be integrated into suitable mitigation strategies. New challenges in wildfire management are emerging, prompting the scientific community to respond with integrative approaches and innovative modeling applications, fostered by technological improvements in computation capabilities and an increase in data availability [7].

Wildfire risk modeling is a key approach to uncover the complex relationships between fire occurrence, its driving factors and potential impacts and is relevant for different wildfire phases: prevention, suppression, and recovery. The methodologies used for risk assessment and modeling have evolved over time, and nowadays, most wildfire risk modeling approaches reflect a broad conceptual framework, which includes several distinct but interconnected components that can be analyzed separately [8] and incorporated into decision-making processes at different levels, such as landscape management [9] or household protection and evacuation [10] (see Figure 1).

This paper presents a concise review of the methods and approaches applied to model wildfire risk in the last few years, focusing on scientific and technical developments since 2018. We describe the tools used to assess the key risk components, how they are combined and applied, and how they contribute to develop suitable products for wildfire management.

Wildfire likelihood and hazard reflect the characteristics of the fire process, responding to specific questions: (1) how it starts, (2) where it happens, (3) where it occurs, and (4) how it will grow. Wildfire occurrence modeling and fire spread simulations have been implemented in different territorial contexts. The former are covered mainly in probability analysis and susceptibility assessments (1, 2, and 3), whereas the latter is used in building scenarios of fire ignition, intensity, and propagation (4). Regression models, easy to use and interpret, have been extensively applied [11,12]. The approaches used to model specifically human-caused fires have been previously reviewed by Costafreda-Aumedes et al. [7] who pointed out that the selection of predictor variables is not based on the available data and on the temporal and spatial scales of the analysis. Where models cover large geographical areas and in coarser nonstationary patterns of wildfire drivers, geographically weighted regression has been used [13,14], and generalized additive regression has also been tested [15].

In the last decade, wildfire modeling has evolved to use advanced machine learning approaches based on algorithms that can improve automatically through experience [16]. Their use presents some advantages because they do not require a prior knowledge of the phenomenon, and they can be used to nonparametric data, being able to identify nonlinear patterns from large data sets. On the other hand, they use complex and sophisticated algorithms, and the interpretation of the importance of the independent variables is not straightforward as in the traditional regression methods [17,18]. The use of such methods, among which Artificial Neural Networks, Random Forests, or Support Vector Machines, is becoming increasingly frequent [19, 20, 21, 22], occasionally focusing on comparative evaluations of different methods [16,23,24]. The application of different machine learning methods in wildfire science has been recently reviewed by Jain et al. [16]. Another approach gaining relevance is Maximum Entropy (MaxEnt), an algorithm that joins the predictive capability of machine learning methods with the explicative ability of traditional regression models, especially used to predict species and ecological niche modeling and lately applied to wildfires in several regions [25, 26, 27].

The response to how wildfires will grow (4) has been increasingly relying on simulation approaches to quantitatively estimate wildfire likelihood and impacts. Different fire simulation systems are available, such as FARSITE, Elmfo, FlamMap, or Burn-P3 [28,29], used to simulate fire ignition and spread in different scenarios of fuel and weather conditions, considering potential landscape interventions, such as fuel treatments and forest management levels [30, 31, 32], in a critical environment combined with remote sensing data [33,34]. These results can support operational decisions, regarding fuel treatment locations, surveillance and suppression activities, and mitigation measures [35,36]. Growing simulation models with Geographic Information Systems (GIS) has been a key issue, and it can be done at three levels: loose, moderate, or tight/close coupling. The loose level uses GIS only as a preprocessing tool, information provider, and a visualizer of results; whereas moderate coupling implies remote procedure calls among GIS and simulation models, such as agent-based modeling approaches [37]. The close coupling simultaneously joins operations of both systems, and it has been developed mainly with the use of cellular automata models within a GIS environment [38].

Wildfire exposure evaluates which assets, and to what extent, are located in fire-prone areas [28]. It results from the analysis of fire occurrence, likelihood, or hazard in relation to highly valued resources and assets (HVRAs), using historical fire data [39] or by coupling stochastic and probabilistic wildfire simulations with the spatial distribution of people and assets, such as dwellings, built-up areas, agricultural lands, or protected areas [40, 41••, 42]. The combination of simulation outputs with GIS has led to models of wildfire transmission to human communities based on the concept of firesheds; the areas with potential to expose communities to fires, according to ignition probabilities and propagation patterns. Using this concept, wildfire exposure was estimated in Spain by combining simulation outputs with spatial layers representing residential areas, built-up structures, and land cover types [43]; land tenure types were mapped, and their relationship with wildfire transmission and exposure levels in Greece was assessed [44]; and wildfire exposure and firehsed extent were obtained in the United States, integrating as well other geospatial attributes, such as travel cost and distance between urban centers and wildland-urban interface areas [45].

Vulnerability represents the potential for loss as a result of wildfires [46]. As a measure of potential wildfire impacts, it depends on the conditions of the wildfire and on the characteristics of the affected assets. When measuring the vulnerability of forests and natural areas, models focus on ecological impacts, simulating the effects of burnt area, heat release, and other fire behavior parameters in the biophysical attributes, mortality rate, or survival response of species in particular ecosystems, such as Alpine forests in France [47] and mountainous grasslands in South Africa [48], or by assessing successional trajectories in Mediterranean pine forests in Spain, via sensitivity analysis and a persistence index [49].

In parallel, many studies in the past decade emphasized the social component to assess the vulnerability of human communities to wildfires, following the research first developed by Cutter et al. [50] with the Social Vulnerability Index. Social attributes of people exposed to wildfires are weighted, aggregated, and mapped based on geospatial or cluster analysis [32,51]. Other recent works have coupled social data with wildfire simulations [40,52], showing which human communities have more or less capacity to absorb the effects and to recover from the impacts of wildfires.

Lately, particular attention has been given to the implementation of self-protection and mitigation strategies by people exposed to wildfires [53]. Models of alternative actions and behavior choices that can modify the response capacity to wildfire events have emerged, assessing evacuation, local sheltering, or stay-at-home options [54, 55, 56]. By applying the Protective Action Decision Model (PADM), by Lindell & Perry [57], a decision-making model that integrates environmental and social cues within a framework of information flows, several studies have evaluated people's behavioral choices in Australia and in the United States, for example, as reviewed by McLennan et al. [58]. Other modeling approaches have been tested, with new methods or coupled with PADM, generally divided into two categories: conceptual models, which develop frameworks to explain people's behavioral steps when deciding what to do in a wildfire emergency; and engineering models, which assess the factors affecting human behavior (choice models) or simulate evacuation travel conditions (traffic models) [59]. For example, a mathematical approach to create a Wildfire Decision Model has been recently proposed, incorporating human behavior assumptions with simulations of people's actions using a binary logit model [60]. In other cases, geospatial tools are used to estimate wildfire and traffic simulations to assess evacuation triggers [60]. Other studies have been focusing on assessing how the characteristics, experience, and social context of exposed communities can influence mitigation and adaptation actions, using interactional approaches to document and compare different combinations of local characteristics. These can allow identifying tailored strategies for wildfire recovery and adaptation, according to the capabilities and needs of distinct human communities [61,62].

Wildfire risk assessment intends to provide an integrated view of fire likelihood and consequences [63]. The relative contribution of each component previously described, is evaluated, then aggregated and summarized in quantitative indices or categories, providing a rating of wildfire occurrence and potential impacts, and further combined with mapping and zoning tools to identify priority areas for intervention. In Portugal, a composite risk index in five categories, which combined burn probability, exposure, and social vulnerability, was applied to the villages of a civil parish [32]. In Australia, a Bayesian decision network model was applied to specifically assess the efficiency of prescribed burning strategies in wildfire risk levels regarding valued assets [9]. In Brazil, a fire risk map was obtained by combining historical fire data with a Kernel density estimator to determine critical areas [64]. In the United States, a multi-criteria decision support model was used to assess fire risk variations in view of potential suppression activities at the operational level [65]. In Canada, wildfire risk has been calculated based on a specific equation that sums probability linked to a range of fire behavior conditions, multiplied by the difference between potential benefits and losses [28]. A European initiative has also proposed a composite risk index that aggregates three dimensions: hazard and exposure, vulnerability, and lack of coping capacity, applicable at different spatial scales and for multiple hazards [66]. Because of the complexity of the phenomenon and the numerous approaches available, the comparison and interpretation of wildfire risk models is still challenging. Also, risk ratings may cover only part of the components, varying in data and methods, weighting, type of outputs obtained, or scale of application [8]. Despite this diversity, common ground among risk models is based on their main purpose of obtaining a grading system for the wildfire threat and on their spatially explicit dimension, to which geospatial tools are particularly suitable [67].

Section snippets

Final remarks

Wildfire science is continuously evolving to respond to environmental changes and societal needs. Currently, we are facing a paradigm shift regarding the development of sustainable planning and operational frameworks, with the aim of reducing fire-induced damages and losses, instead of focusing on burned area reduction [6]. Risk models are developed with the ultimate goal to support wildfire management and decision-making at multiple levels. The need to integrate the scientific knowledge and ...

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. ...

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Assessment of social vulnerability to forest fire and hazardous facilities in Germany
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... International studies on wildfire risk have a long tradition in the United States and Australia [36,37]. Scientists developed fire risk indices for wildland-urban or rural-urban interfaces [38–25]. Fire risk assessment methods developed in the context of fires in urban settlements include "fixed-mechanics-based or other physics-based models to portray fire spread", risk models, "empirical, retrospective, models of house loss versus distance from the urban edge", or other linkages calculated post hoc [24,...

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